Design Pattern Analysis in a Next.js-Based Video Streaming Application

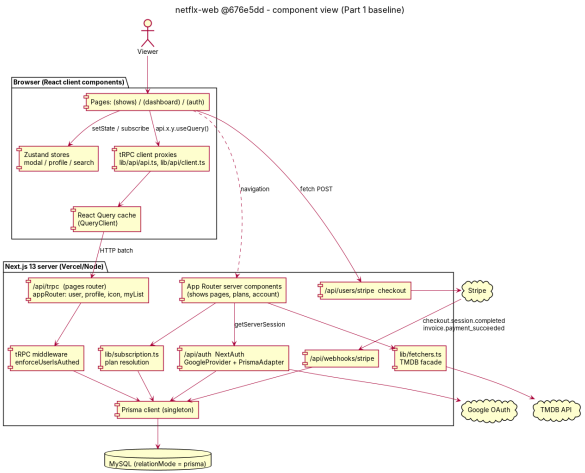
23CSE455 Design Patterns — Case Study, Team A12

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Application under study

- **netflx-web** (sadmann7), Netflix clone
- Next.js 13.4 app router, tRPC 10, NextAuth 4, Prisma 4 (MySQL), Stripe, Zustand, React Query, TMDB API
- Baseline commit `676e5dd` (2024-06-12)
- No LICENSE file — academic analysis only
- Seven areas: user management, catalogue, subscription, recommendation, authentication, content access, caching



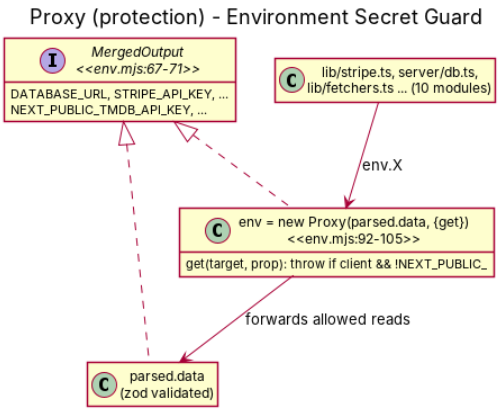
Part 1 — pattern catalogue at the baseline

Pattern	Instances	#	Concrete
Chain of Responsibility	Authenticated Procedure Pipeline	1	1
Singleton	Shared Prisma Client; Stripe API Client	2	2
Observer	Show Modal Selection; Active Profile Selection; Search Results	3	15
Decorator	Persisted Profile Store (devtools(persist(...)))	1	3
Adapter	NextAuth Prisma Persistence Adapter	1	1
Strategy	Sign-in Provider Strategy	1	1
Facade	TMDB Catalogue Facade	1	1
Composite	API Router Tree	1	5
Proxy	2 tRPC client proxies; Environment Secret Guard	3	3
Read-Through Cache	My List Query Cache	1	1
10 types		**15**	**33**

6 rejected (e.g. dead caching proxy, no recommendation logic), 4 uncertain (e.g. Stripe webhook).
Every range checked by script.

Example: Environment Secret Guard (protection proxy)

- `env = new Proxy(parsed.data, {get})` in `env.mjs:92-105`
- On the client, any non-`NEXT_PUBLIC_` read throws
- Found on the second review pass (LLM's first answer missed it)
- Explains a smell found in Part 2: plan data was duplicated because the server list reads secrets



Part 2 — three changes

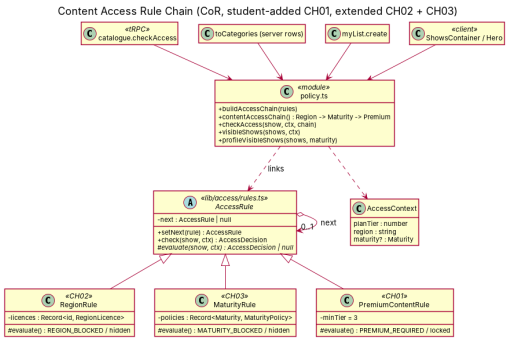
Change	What	Tests
CH01 premium + Ultra plan	tier-based premium titles, Ultra (7 profiles), subscription middleware	29/29
CH02 regional restrictions	licences per country, geo header, server-side row filtering	45/45
CH03 parental control	KIDS / TEEN / ADULT rating per profile	65/65

- Characterization tests written first and run on the *original* code (19/19)
- New student-added **Content Access Rule Chain**: Region → Maturity → Premium (hide before lock)
- Existing tRPC middleware chain extended with `attachSubscription`

Pattern evolution

Snapshot	Types	Inst.	Concr.
baseline	10	15	33
CH01	11	17	37
CH02	11	17	39
CH03	11	17	42

- New: Content Access Rule Chain (CoR), Access Rule Evaluation Template (Template Method)
- Extended: auth pipeline, router tree, profile observers
- Modified: TMDB facade, search observer; none removed



Refactoring — real smells, measured with the same tool

Smell	Technique	Before	After
Duplicated plan data (UI vs server, already drifted)	shared client-safe catalogue	139 NLOC	123 (+1 plan)
Copy-pasted TMDB fetch blocks	rows as data + `fetchRows`	max CCN 9	3
Same row array in 3 pages	`GENRE_ROWS` + `toCategories`	174 NLOC	116
Duplicated zod schema / lookups in My List	Extract Variable / Function	169 NLOC	114

Whole `src/`: 6694 → 6916 NLOC (+3.3%), avg CCN 2.31 → 2.26. LOC alone does not prove better design; the evidence is behaviour preserved by tests.

Part 3 — four languages, one shared fixture

	Java	Python	JavaScript	C++
cases passed (10 problems)	107/107	107/107	107/107	107/107
total source LOC	497	316	316	558
max function CC	5	6	5	6
median process time (ms)	86.7	14.9	32.6	1.8

- Proxy: JS `Proxy` trap ≈ source; Python `__getattr__`; Java `reflect.Proxy`; C++ plain class
- JS needs no classes for Observer / Decorator / Adapter (functions and object literals, like zustand and next-auth)
- Times are start-up dominated; `dataclasses` import costs ≈20–26 ms in Python

LLM use and verification

- One Claude (Opus 5.5) session for Parts 1–3; prompts, outputs and claim tables in `Part*/llm/`, summary in `data/llm.csv`
- Errors found by checking: missed instance (env proxy), 3 wrong line ranges, NLOC first measured without `.tsx`, a max-CCN rise that was a lizard artefact
- Not accepted: Stripe webhook as Observer (kept uncertain)
- All generated code is covered by tests (65 app tests, 428 cross-language checks)

Findings

- Most patterns are inherited from frameworks (tRPC, zustand, NextAuth, React Query) and configured, not written
- Recommendation has no pattern because it has no logic
- A chain of rules absorbed premium, region and parental control with one class each
- Pre-existing issues: client-side PIN check, missing ownership checks on profile / My List, dead caching proxy, two parallel tRPC clients
- Limits: no end-to-end run with real Stripe / TMDB / Google; sample licence data